

1. Title

FreightQuote AI

Agentic Maritime Brokerage & Freight Quotation Platform



2. Project Statement

Global maritime freight brokerage involves complex decision-making across shipping routes, freight pricing, weather conditions, customs regulations, and profit margin calculations. Freight brokers often rely on manual processes, spreadsheets, and fragmented information sources, resulting in delayed quotations, pricing inaccuracies, compliance risks, and reduced profitability.

This project builds an Agentic AI-powered Maritime Brokerage Platform that automates freight quotation generation through a network of specialized AI agents. These agents collaborate to identify optimal shipping routes, calculate freight pricing, assess weather risks, validate customs requirements, and optimize profit margins.

The platform enables shipping companies, freight forwarders, and logistics brokers to generate accurate quotations faster, reduce operational risk, improve customer responsiveness, and maximize profitability.

3. Outcomes

- Automated freight quotation generation.
- Intelligent shipping route optimization.
- Dynamic freight pricing recommendations.
- Weather-aware route risk assessment.
- Customs compliance validation.
- Margin optimization and profitability analysis.
- Real-time shipping intelligence dashboard.
- Automated broker recommendations.
- Improved quote turnaround time.

- Enhanced brokerage profitability.
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4. Modules to be Implemented

1. Route Agent

- Identify optimal shipping routes.
 - Analyze transit times.
 - Compare carrier options.
 - Evaluate port congestion.
 - Recommend best route alternatives.
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2. Pricing Agent

- Calculate freight rates.
 - Analyze carrier pricing.
 - Estimate shipping costs.
 - Apply dynamic pricing adjustments.
 - Generate customer quotations.
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3. Weather Agent

- Monitor weather conditions.
 - Identify route disruptions.
 - Assess storm risks.
 - Predict weather-related delays.
 - Recommend alternative routes.
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4. Customs Agent

- Validate import/export regulations.
 - Analyze customs requirements.
 - Verify documentation requirements.
 - Identify compliance risks.
 - Generate customs readiness reports.
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5. Margin Agent

- Calculate profitability.
 - Optimize quote margins.
 - Analyze operational costs.
 - Generate pricing recommendations.
 - Track brokerage performance.
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6. Freight Intelligence Engine

- Aggregate findings from all agents.
 - Generate final freight quotations.
 - Calculate shipment risk scores.
 - Produce operational insights.
 - Coordinate agent decisions.
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7. Dashboard & Reporting Module

- Freight Quotation Dashboard
 - Route Intelligence Dashboard
 - Customs Dashboard
 - Risk Dashboard
 - Executive Brokerage Dashboard
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8. Notification & Workflow Module

- Quote Approval Workflow
 - Customer Notifications
 - Email Alerts
 - Teams/Slack Notifications
 - Shipment Risk Alerts
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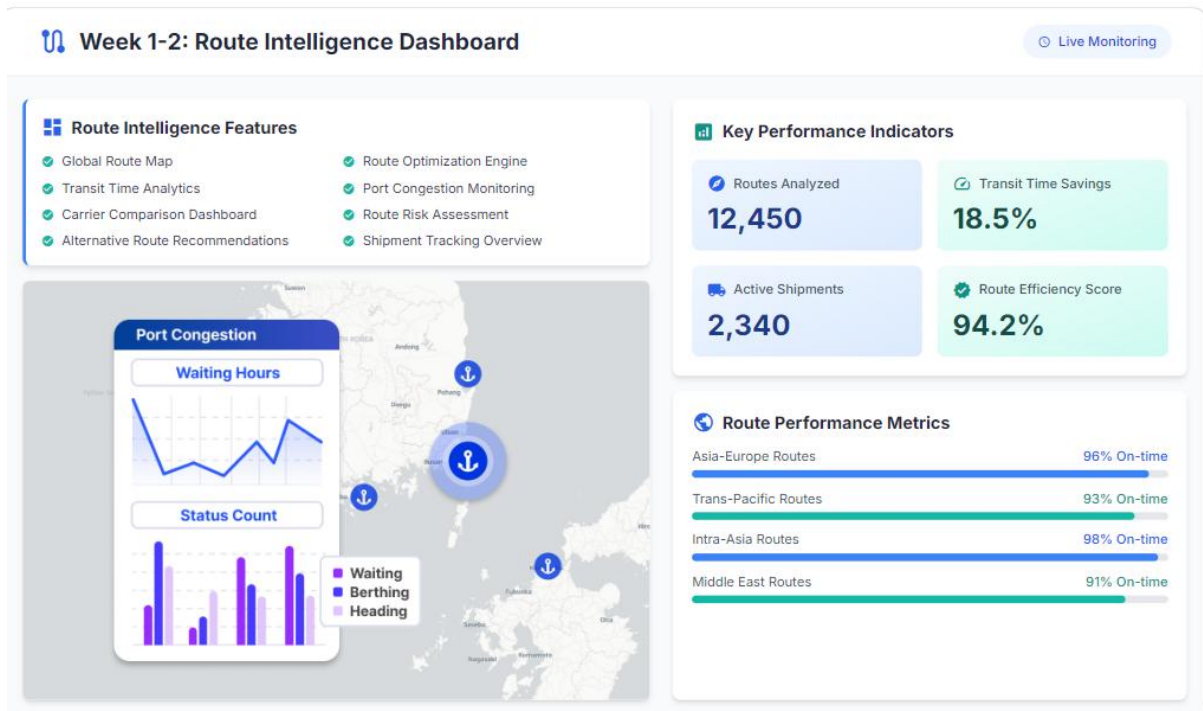
5. Week-wise Module Implementation

Milestone 1 (Weeks 1–2)

Route Intelligence & Quotation Foundation

- Build Route Agent.
- Integrate shipping route datasets.

- Develop transit time estimation.
- Generate route recommendations.
- Create quotation dashboard.



Milestone 2 (Weeks 3–4)

Pricing Intelligence & Margin Optimization

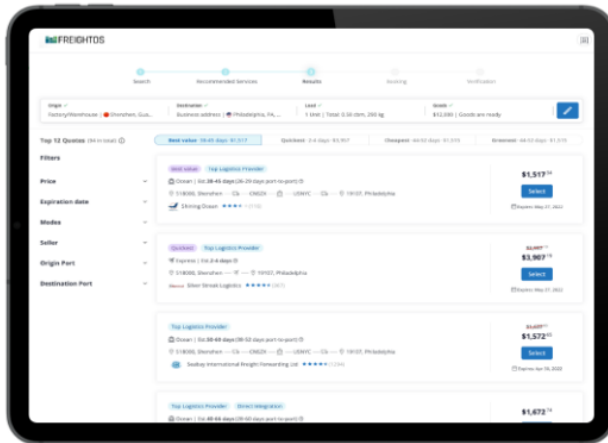
- Build Pricing Agent.
- Build Margin Agent.
- Develop freight pricing engine.
- Generate dynamic quotations.
- Optimize brokerage margins.

Week 3-4: Pricing & Margin Intelligence Dashboard

AI-Powered Pricing

Pricing Intelligence Features

- Freight Pricing Engine
- Dynamic Pricing Recommendations
- Quote Generation Panel
- Profitability Analytics
- Carrier Rate Comparison
- Brokerage Margin Calculator
- Cost Breakdown Analysis
- Customer Quote Management



Key Performance Indicators

Quotes Generated

8,450

Average Margin

14.8%

Revenue Forecast

\$12.3M

Pricing Accuracy

96.7%

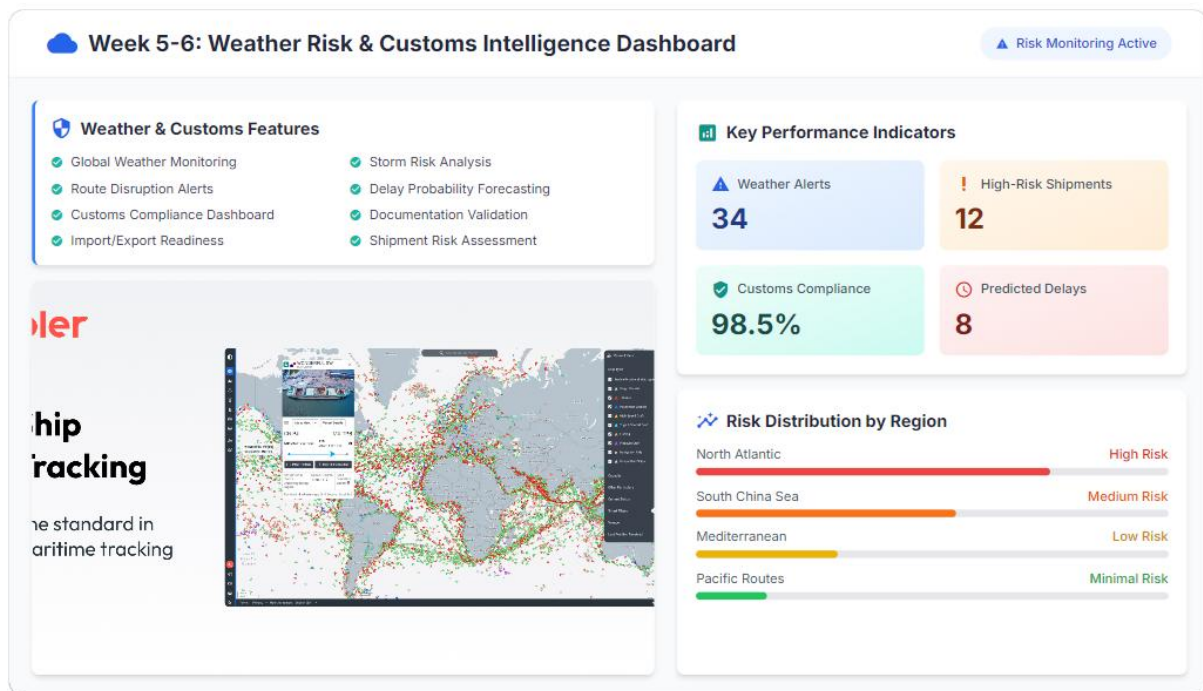
Margin Distribution by Trade Lane



Milestone 3 (Weeks 5–6)

Weather Risk & Customs Intelligence

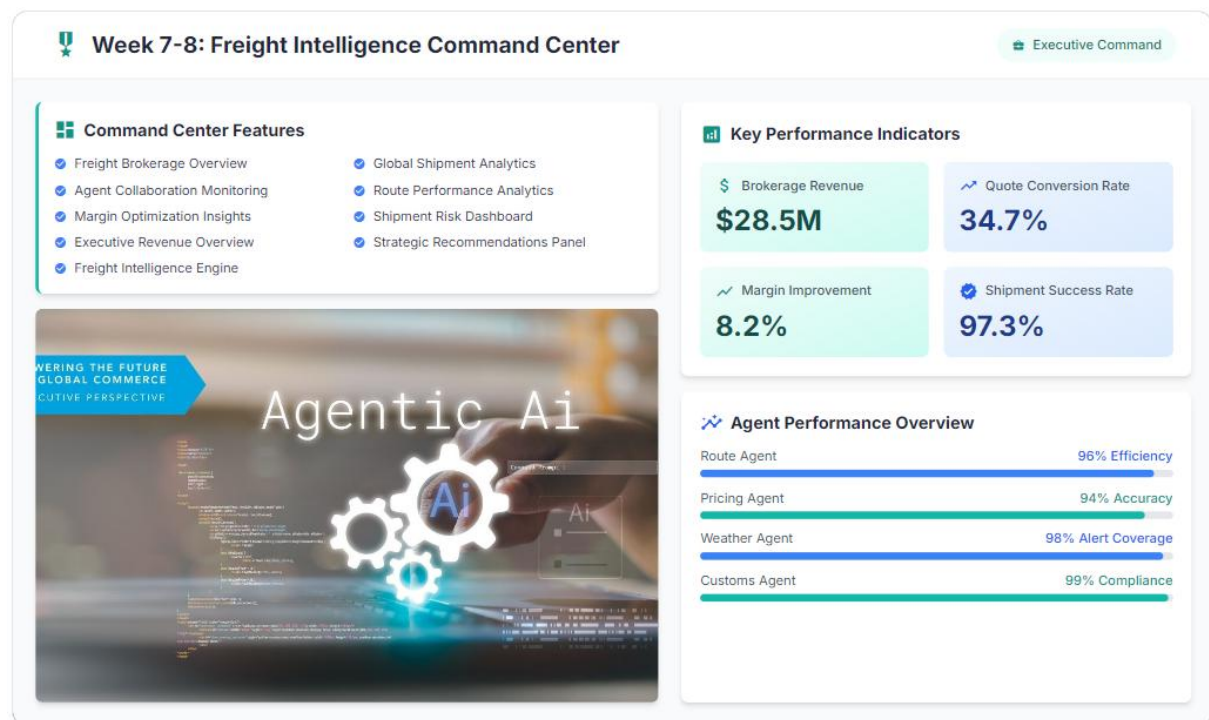
- Build Weather Agent.
- Build Customs Agent.
- Develop weather risk assessment.
- Implement customs validation workflows.
- Generate shipment risk reports.



Milestone 4 (Weeks 7–8)

Freight Intelligence & Enterprise Deployment

- Build Freight Intelligence Engine.
- Develop executive dashboards.
- Implement agent orchestration.
- Perform end-to-end testing.
- Deploy production-ready platform.



6. Evaluation Criteria

Milestone 1 (Week 2)

- Route recommendations generated successfully.
- Transit time estimation operational.
- Quotation dashboard functional.

Milestone 2 (Week 4)

- Freight pricing engine operational.
- Margin optimization functioning.
- Dynamic quote generation successful.

Milestone 3 (Week 6)

- Weather risk analysis operational.
- Customs validation functioning.
- Shipment risk scoring generated.

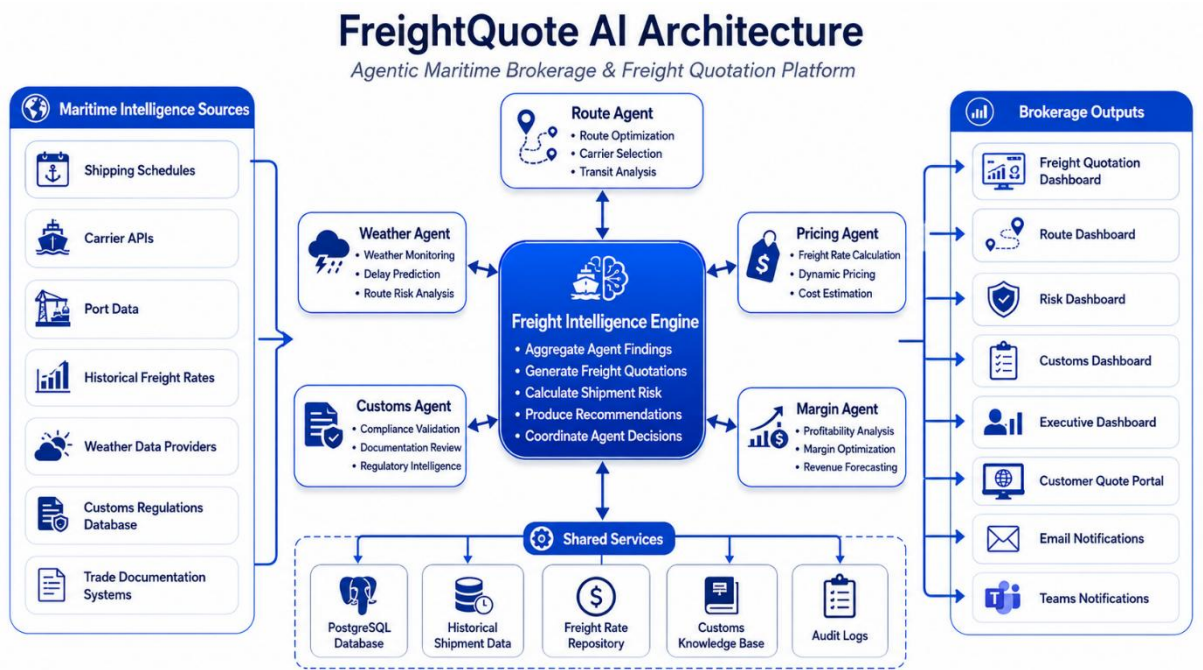
Milestone 4 (Week 8)

- End-to-end platform deployed.
- Freight Intelligence Engine operational.
- Executive dashboards available.
- Automated quotation generation supported.

7. Workflow Diagram



8. Architecture Diagram



9. Database Schema

FreightQuote AI Database Schema

